

Climate Resiliency Through Data: Managing Stormwater Sewers in Detroit

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ABSTRACT

Climate change is intensifying weather around the world. In cities like Detroit, USA, larger storms are resulting in widespread flooding and sewer overflows. To adapt to the changing climate, Detroit is modernizing its sewer infrastructure, adding sensors, robotics, and advanced algorithms that have the potential to increase the capacity and adaptability of its sewer system. The water utility is struggling, however, to incorporate these new technologies into its existing user workflows. We have conducted a user study of water operators in Detroit focused on understanding how they currently visualize and use one of their most critical data sources — weather data. Based on our findings, we have developed a new weather dashboard that minimizes weather data uncertainty by synthesizing multiple sources. This research aims to inform the design of new data interfaces for water operators, and learn best practices for incorporating uncertain data in data-driven decision-making processes.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

climate resilience, civic technology, cyber-physical systems, water infrastructure, uncertainty

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1 INTRODUCTION

In the summer of 2021, the city of Detroit, USA, experienced three “100 year” storms. These catastrophic weather events resulted in widespread flooding, pollution, and extensive property damage — particularly concentrated in already vulnerable, low-income neighborhoods [10, 12]. As a result of climate change, this pattern of intensifying weather is only going to continue. Now, more than ever, there is an urgent need to improve climate resilience in Detroit, and cities around the world [10].

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In Detroit, hundreds of kilometers of pipes, sewers, and basins have been built to control flooding and convey water away from the city during storms. This infrastructure is anything but static, featuring hundreds of pumps, valves, and gates that are operated in real time [2]. At the helm of this control system are water operators — humans in the loop who use data from weather forecasts and sensors to make decisions about how to route water through the city. One of the biggest challenges for these operators is the inherent uncertainty of weather forecasts, which makes it difficult to prepare for impending floods. As a result, the most effective operators bring a lot of specialized knowledge and experience to their jobs.

The city of Detroit is now planning to improve its flood resilience by introducing automation and smart technologies to the sewer system. Part of this process is the desire to modernize operator decision making with more advanced methods such as machine learning-based recommendations [2]. This is in line with the broader global movement of civic organizations engaging more deeply with Internet of Things (IoT) technologies [1]. Similar to other civic organizations, however, the Detroit water utility is experiencing difficulties integrating these new technologies into existing user workflows.

In this research project, we have conducted a user study of water operators in Detroit, focused on understanding how they currently use data to make decisions and why smart technologies have not already percolated into this space. This case study explores a specific instance of how design thinking can aid the integration of smart technology into a civic space — water infrastructure management. The findings of this study, however, may be more broadly applicable to the field of civic technology as a whole. This research aims to inform the design of the next generation of water infrastructure user interfaces. Modernizing water operators’ decision-making processes has the potential to greatly increase the storm sewer infrastructure system’s capacity and resilience, preparing Detroit for the next century of storms [3, 6].

2 USER INTERVIEWS

We interviewed eight groups of water operators performing a variety of roles in the Detroit sewer network, ranging from managing maintenance to operating water treatment facilities. We asked operators to describe how they currently read the weather forecast and discussed which data visualization features they liked and disliked. We asked open-ended questions about how operators use weather data before and after a storm, and what changes could be made to improve the usability of their current information systems.

Broadly, water operators use weather data for two reasons: (1) to determine staffing needs before an upcoming storm, and (2) to estimate the rate of water flowing into their system during a storm. The common goal of operators is to dewater, or remove



Figure 1: Flooding damages critical infrastructure, personal property, and the environment. Photographs by DVIDSHUB, U. S. Geological Survey, Mrs. Gemstone, and David Shankbone. [Public domain], via Flickr. [4, 7, 9, 11].

water from their system, as fast as possible, which minimizes the risk of flooding in their region. Regulations limit the rate at which upstream operators can dewater in order to protect the downstream areas of Detroit from flooding [2]. Operators use the precipitation forecast, as well as other weather metrics, to estimate the flow of water into their system, which informs when and how fast they need to dewater.

Currently, operators use a variety of weather data sources to learn the forecast. The regional utility has multiple contracts with proprietary data sources to provide customized dashboards for their operators. Which of these tools operators use, however, is left up to them. The operators interviewed look at between 2 and 5 different sources, including proprietary dashboards, news sites, phone apps, meteorological reports, and storm radar maps. Operators reported checking the forecast as little as once a day when no rain is predicted, and as often as every fifteen minutes in the lead up to a storm.

Among the sub-groups interviewed, none reported any formal guidelines in place for how they should interpret weather data. Instead, operators described rules of thumb, such as *“a third, a fourth of an inch of rain is the threshold. We used to come in whenever it was at least a 50% chance of rain, but now we’re more cautious.”* (P6). Five operator groups reported regularly checking other weather metrics such as temperature and wind speed to gain a deeper understanding of how water will move during the storm event: *“Between the wind patterns and the graph from NOAA’s website, that’s pretty much what I make my decision based off of.”* (P4). Operators then compare weather data against sensors embedded in their infrastructure systems to make decisions on how to operate their plants.

3 RESULTS

Although operators have a variety of weather dashboards to choose from, none of the existing tools compile all the data they need in one place. In part, this is because water operators seek to reduce risk in decision making by looking at how different weather forecasts compare. Although different weather dashboards all predict the same variable — expected rainfall in Detroit — there are often variations in the predictions, stemming from the uncertainty in the underlying measurements. Operators understand this and purposefully look at multiple forecasts to mitigate the uncertainty — *“I wouldn’t put my eggs in one basket, especially when words like torrential are getting thrown around. I would go and check as many different sources as possible.”* (P5).

Acquiring this redundancy in data takes time, and with Detroit’s recent track record of flooding, operators are spending more time looking at the weather forecast to confirm they haven’t missed anything. When asked at what point they felt that they had enough information about the weather, one operator group responded, *“No, I’m always looking for more.”* (P4). Inefficiencies in the existing data communication pipeline are likely to have a tangible detrimental impact on the sewer system. Time operators spend finding and interpreting weather data is time that could be better spent monitoring their infrastructure systems and communicating with other operators.

All eight operator groups expressed that they want as many data as they can get, and five operator groups reported being unsatisfied with the resolution and/or accuracy of data they currently have. Simultaneously, six of the operator groups noted that due to the inherent uncertainty in the weather forecast, they prefer to see generalized summaries of weather data that indicate a confidence interval or a range for the expected precipitation, before looking at more detailed reports. This is consistent with findings from other studies of users working with uncertain weather forecast data [5]. The typical operator workflow is summarized in Figure 2 (blue). Based on the interview findings, we have developed suggestions (orange) for improving this process. Displaying data from multiple sources, side by side, could reduce the time operators spend searching for redundant data and facilitate comparison between different forecasts.

There are no tools in use that translate weather forecast data into expected water inflow, and as a result, the learning curve for new operators can be steep — *“People here have to be highly specialized, and that’s a problem. If they get sick, if they go [to another job], it’s hard to replace them.”* (P8). This is an area in which smart technologies could improve the existing user workflow [8]. Developing predictive water flow forecasting and translating these forecasts into recommendations for operators could reduce the system’s reliance on individual users’ experience. Figure 3 illustrates an example of how this could be beneficial. A smart recommendation system prevents flooding at a downstream location B, by suggesting that upstream location A hold back water until the storm has passed, allowing B to dewater first. Smart recommendation systems such as this have been prototyped, but have yet to be adopted in the real world [3].

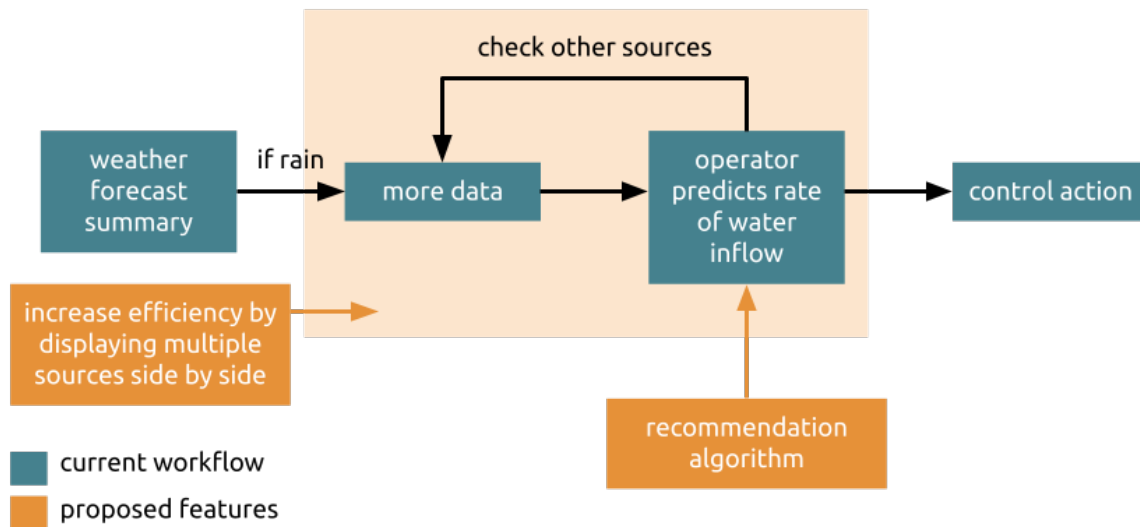


Figure 2: A summarized operator workflow (blue) shows how data is used to make decisions. Proposed features to improve the existing process are shown in orange.

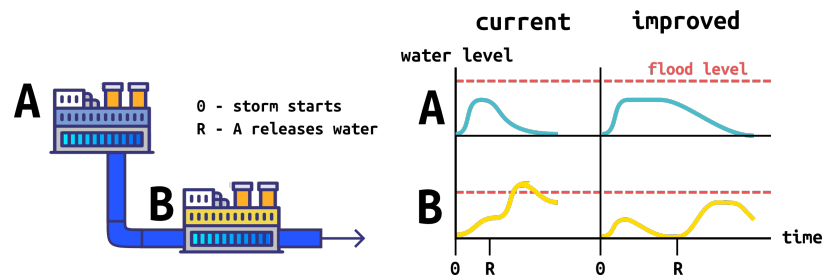


Figure 3: Example scenario showing how the use of a smart recommendation algorithm could prevent flooding by encouraging water operators to work together.

Operators are cautious about using more advanced modeling tools and smart algorithms. Of the eight operator groups interviewed, six were candidates for using new modeling tools such as flow forecasting. Of these six, two were skeptical of the utility they could get out of flow forecasting, and three had mixed or ambivalent expectations. Only one felt that flow forecasting would be very useful. The concerns raised mainly involved the potential accuracy of these predictive tools. “We’re going to find difficulty getting away from real-time decision making, you don’t know what you have until it’s on your doorstep.” (P4). Operators understand that the uncertainty inherent in the input to these algorithms will be reflected in the output. Making a bad control decision during a storm can result in flooding, pollution, and massive fines for the water operator groups. As a result, moving away from existing water management methods and relying on new, less-established tools is seen as too risky by the majority of operators.

4 A NEW WEATHER DASHBOARD

Based on interview findings, we have designed a new weather forecast dashboard (Figure 4) that synthesizes data from multiple sources to facilitate comparison and mitigate the impacts of uncertainty. This dashboard displays the data identified as most important by operators. At the top of the dashboard is a summarized rainfall forecast for the next five days. As users scroll down, they can find more detailed weather forecast information, including an interactive predicted rainfall map and longer-term forecasts spanning the next two weeks. Figures are enlarged when clicked on, and the source of data is listed along with a link to the original website. In short, this dashboard allows operators to view all relevant data on one single-page website.

In a second, ongoing round of interviews, we are using this dashboard as a high-fidelity functional prototype to gain deeper insight into how water operators use this data. Our weather dashboard did not change the visual aspect of the original data presentation and as a result, the interviewed operators all quickly understood how to interpret the data. Out of the four interviews conducted so far, two

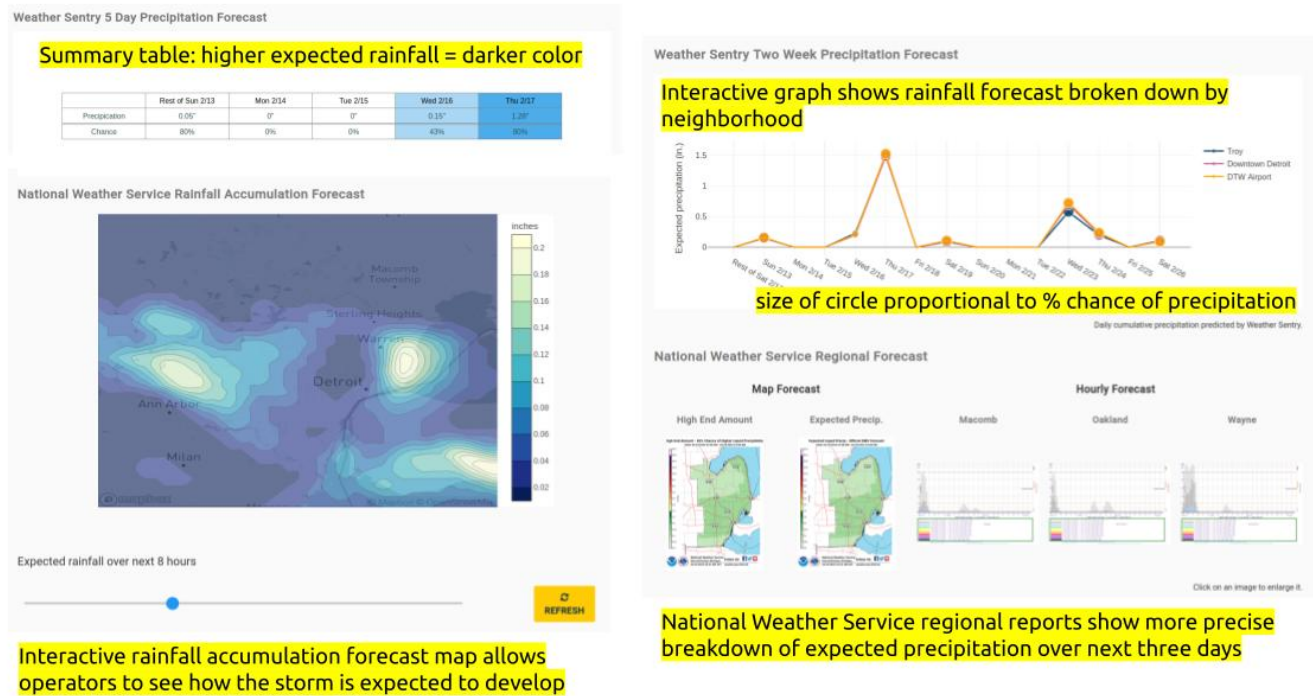


Figure 4: Select screen captures of our weather forecast dashboard for stormwater operators.

operator groups suggested adding higher resolution precipitation forecasts. Operators also expressed enthusiasm for the predicted rainfall map, and two operator groups suggested expanding this feature.

All of the interviewed operators reported comparing weather forecast data from multiple sources in their daily workflow, and in the second round of the interviews, all operators agreed that this was a useful feature of the dashboard. “Since you’ve included multiple data sources, that definitely helps us go, yeah they all seem to be saying the same thing.” (P2). Operators were satisfied with the summary of data presented; when asked about its trustworthiness, all four agreed that because they understood the source of the data, they felt they could trust the dashboard. “I would trust [this dashboard] just as much as anything else I’m using right now, or any news broadcast or other site I’m using.” (P2).

5 DISCUSSION AND FUTURE WORK

When working with uncertain data, comparing multiple sources, providing a summary and showing uncertainty bounds are useful strategies for enhancing user understanding and trust. Showing where data come from and how data are being transformed seems to be a particularly valuable strategy when introducing users to new data sources. In other studies on the integration of technology into civic space, it has been noted that simply collecting data is often not enough, civic organizations need to learn how to use the data to make good decisions [1]. Contextualizing new data sources by explaining where they come from may be one way to speed up the learning process. In the water operator case study, one way this

could be done is by displaying flow forecasting data alongside the raw weather forecast data from which it is derived.

Smart technologies have the potential to transform urban life for the better, but effectively communicating novel data to decision makers remains an understudied problem. In future work, we will explore adding further data synthesis tools to our weather dashboard, such as a panel directly comparing the precipitation forecast from multiple sites. We will also work towards adding more advanced modeling data to the dashboard, displayed alongside the primary weather prediction, to investigate whether this is an effective way to build user trust and encourage the adoption of these smart technologies.

6 CONCLUSION

Our findings suggest that comparison of multiple sources is an effective tool for mitigating uncertainty in data, and that presenting data from novel sources alongside already-trusted data may be one way of encouraging the adoption of smart technologies. These strategies may have utility in other civic technology spaces, as city planners and urban activists work to incorporate IoT into more aspects of urban life. Modernizing water infrastructure operation by introducing smart technology is expected to improve the performance of the Detroit sewer system, resulting in less flooding, lower rates of pollution, and improved water quality across the region. This is one way Detroit, and cities around the world, can increase their resilience to climate change.

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